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# Library Management System

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# Introduction

- Library Management System is a Java-based application
- It is used to manage books and users in a library
- Helps in storing and retrieving book details easily
- It reduces manual work
- This project automates the process of managing books in a library.

# Objective

- To store book details (ID, Name, Author)
- To perform operations like:
  - Add book
  - View books
  - Search book
  - Issue/Delete book
- The main aim is to simplify library operations using Java.

# Features

- Add new books
- Display all books
- Search book by ID or name
- Issue/Delete a book
- These features are similar to real library systems.

# Tools Used

- Java
- JDK
- IDE (Eclipse / IntelliJ)



# System Design

- Uses **ArrayList** to store book objects
- Each book is represented as an object
- Methods are used for operations
- ArrayList helps to dynamically manage book records.

# Class Structure

- **Book Class**
- id
- name
- author
- **Main Class**
- addBook()
- displayBooks()
- searchBook()
- deleteBook()
- The program is divided into classes to maintain structure.

# Code Logic (Simple Explanation)

- Book class → stores book data
- Constructor → initializes values
- ArrayList → stores multiple books
- Methods → perform operations
- We use object-oriented programming concepts here.



# Code

```
import java.util.*;
class Book {
    int id;
    String name;
    String author;
    Book(int id, String name, String author) {
        this.id = id;
        this.name = name;
        this.author = author;
    }
}
public class LibrarySystem {
    static ArrayList<Book> list = new ArrayList<>();
    public static void addBook(int id, String name, String author) {
        list.add(new Book(id, name, author));
        System.out.println("Book Added!");
    }
    public static void displayBooks() {
        System.out.println("\n--- Book List ---");
        for (Book b : list) {
```

```
if (b.id == id) {  
    System.out.println("Found: " + b.name + " by " + b.author);  
    return;  
}  
}  
System.out.println("Book not found");  
}
```

```
// Delete Book  
public static void deleteBook(int id) {  
    list.removeIf(b -> b.id == id);  
    System.out.println("Book Deleted!");  
}
```

```
public static void main(String[] args) {
```

```
    // Adding Books  
    addBook(1, "Java Basics", "James Gosling");  
    addBook(2, "Data Structures", "Mark Allen");
```

```
    // Display Books  
    displayBooks();
```

```
    // Search Book  
    System.out.println("\n--- Search Result ---");  
    searchBook(1);
```

```
    // Delete Book  
    deleteBook(2);
```

```
    // Display After Deletion  
    System.out.println("\n--- After Deletion ---");  
    displayBooks();
```

```
    }  
}
```

# Output

```
2  Data Structures  Mark Allen
```

```
-- Search Result ---
```

```
Found: Java Basics by James Gosling
```

```
Book Deleted!
```

```
-- After Deletion ---
```

```
1  Java Basics  James Gosling
```

# Conclusion

- The Library Management System is a simple and effective Java application
- It demonstrates:
- Object-Oriented Programming
- Data handling using ArrayList
- Method-based operations
- The system:
- Reduces manual work
- Improves efficiency
- Organizes book records properly
- It is:
- Easy to use
- Flexible
- Suitable for small libraries